Honors Calculus III

Syllabus

Spring 2026 **Math 16300**

§A. What is this class?

Math 16300 is the third and final quarter of the Honors Calculus sequence. In Math 16100–16200, you developed a rigorous theory of limits, continuity, differentiation, and integration for functions of a single real variable. This quarter, we push further. What does it mean to add up infinitely many numbers, and does the result depend on the order? What does it mean to build a function as a limit of simpler functions, and do properties like continuity and integrability survive? What happens to the calculus you have learned when you move from $\mathbb{R}$ to $\mathbb{R}^n$?

The first six weeks address the first two groups of questions, following Chapters 22–24 of Spivak's *Calculus*: we develop the theory of sequences, infinite series, and sequences and series of functions. The final weeks take up the third, lifting our theory of convergence, continuity, and compactness from $\mathbb{R}$ to the setting of metric spaces and ending with a glimpse of point-set topology. This material serves as the entry point to upper-level courses in real analysis and topology.

A.1 How class meetings work

This is an **active, discovery-based course**. You should expect to be *doing mathematics* during most of every class period, not passively watching someone else do it. This is not a fully student-led IBL course (I will lecture when it is the right tool), but even during lectures, you will be actively involved, supplying key steps in proofs and testing ideas before they are formalized. Come prepared to think, write, and talk with your classmates.

Please make use of my office hours and plan to work hard. Begin studying for the quizzes well in advance. Put your good study habits into practice by reviewing class notes and working through extra problems when necessary, so that you can identify weaknesses and seek help. Remember that part of doing real math is productive failure: you'll try things that don't work; learn something from that failure; try something new that works a bit better, and … after a while, you will figure it out, and come out with a much more robust understanding of the structure of mathematics!

§B. Key Information

Class Meetings

MWF 10:30 AM – 11:20 AM, KPTC 101

How to contact me

- **Email:** subhadip@uchicago.edu
- **Office:** Eckhart 120B

Office Hours

See Canvas for up-to-date hours. You can also email me to set up an individual meeting.

Study Materials

- **Lecture notes:** The primary resource for this course. Posted to Canvas before each class meeting.
- **Textbooks (recommended, not required):**
  - Michael Spivak, *Calculus*, 4th edition. Chapters 22–24 closely parallel our first six weeks.
- **Computing and Graphing:** We will occasionally use Desmos for visualization and Mathematica for in-class demonstrations. You can get a free Mathematica license using your CNetID. No prior programming knowledge is required.

Class announcements

- **Available on:** https://canvas.uchicago.edu/courses/71293

  Check Canvas and your UChicago email at least once before and after each class.

§C. Learning Objectives

Upon completion of this course, you will be able to

- **state** definitions and major theorems precisely, including the monotone convergence theorem, the Bolzano–Weierstrass theorem, the Cauchy convergence criterion, Dirichlet's Test, Dini's theorem, the Weierstrass $M-$test, and the Trichotomy theorem;
- **prove** results directly from definitions, demonstrating fluency in ε–N arguments for sequences and series, and $\varepsilon-\delta$ arguments for convergence and continuity in metric spaces;
- **apply** the major theorems to establish new results about sequences, series, and functions, selecting the appropriate tool from the available techniques;
- **distinguish** between superficially similar concepts, such as pointwise and uniform convergence, absolute and conditional convergence, and open and closed sets, and **explain** why the distinctions matter using specific examples;
- **construct** examples and counterexamples with prescribed properties, demonstrating the necessity of hypotheses in the major theorems;
- **determine** whether a proposed mathematical statement is true or false, and **justify** the conclusion with a complete proof or a decisive counterexample.

§D. Using GenAI in Math 163

I do not recommend using generative AI tools (such as ChatGPT, Gemini, or PhoenixAI) on homework for this course. The weekly problem sets are designed to develop your proof-writing skills through practice, and outsourcing that practice to an AI undermines the purpose. That said, I will not penalize you for using AI on homework, provided you:

- cite any AI usage clearly (tool used, what you used it for, what you changed);
- understand and can explain everything you submit.

Outside of homework, you are welcome to use these tools to support your learning, and I expect you to mention any such use in classroom surveys. You might, for example, ask an AI to generate extra practice problems on ε–N arguments, to give you a first-pass explanation of a definition you find confusing, or to help you debug a Mathematica command during an exploration activity. If you do use AI in any of these ways, I expect you to mention it in classroom surveys.

This course is designed to develop skills you will need regardless of what tools are available now or in the future:

- how to ask intelligent questions about mathematical claims,
- how to connect ideas across different parts of the subject, and
- how to communicate a mathematical argument clearly enough so that another mathematician would be convinced of your expertise.

I encourage you to experiment with the tools you are familiar with, but make sure you are building these skills yourself.

No technology of any kind (calculators, phones, AI tools, notes) is permitted during in-class quizzes or the oral final examination.

The mastery standards on quizzes and the oral final are the primary evidence of your mathematical understanding. AI cannot help you there. The homework is where you build the skills that those assessments measure, so use it wisely.

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§E. Components of the course

Our section uses a **mastery-based grading system** (also called standards-based grading). Your final grade is determined by a checklist with three components: how many of the course standards you have demonstrated mastery of, your overall homework score, and your oral final.**There are no point values, no weighted averages, and no curves**. Standards are demonstrated on checkpoint quizzes and through token retesting during office hours; the full list is in section L.

When you submit most work, I will evaluate it relative to the quality standards made clear on each question. If your work meets the standard, then you will receive full credit for it. Otherwise, you will get helpful feedback and, on most items, the chance to reflect on the feedback, revise your work, and then reassess your understanding.

This feedback loop represents and supports the way that people learn. Learning happens over time, as we revisit ideas and reflect on them. In this class, your final grade will reflect how well you **eventually** understand each topic. You can make mistakes without penalty, as long as you **eventually** demonstrate fluency in the topic.

E.1 Attendance and Engagement

In this class, **showing up is necessary but not sufficient**: true engagement means being present, prepared, and actively participating in your own learning and the learning of your peers. Because so much of our class time involves collaborative problem-solving, your participation directly affects not just your own learning but the learning environment for everyone.

Being Present

Please show up to class **on time** and ready to engage. **Attendance will be taken.** If you need to miss class for any reason, email me as soon as possible and make a plan to catch up. In particular, you should:

- Read through the class notes posted to Canvas;
- Review the corresponding content using the textbook or other resources;
- Try the relevant homework problems *before* seeking additional help from me with the material from that day.

Please do not miss class without contacting me. I don't care whether you think it's a good 'excuse' or not. Just send me an email or reply to mine! This isn't about enforcing perfect attendance; it's about keeping an open line of communication so I can support you effectively. Lack of communication about absences may affect your final grade.

Engaging Actively

We'll be learning together, and the participation of every student, whether in whole-class, small-group, or individual settings, creates the best learning environment for everyone. As a college student and future professional, full engagement is not optional; it's expected.

You can demonstrate your engagement in a number of ways:

- Working seriously on in-class exercises, both individually and with your neighbors.
- Giving constructive feedback during in-class discussions.
- Asking relevant questions in class, during office hours, or through email.
- Creating an inclusive and welcoming class environment for peers.

Consistent effort is the baseline, not the bonus. So, please show up, speak up, and support each other.

Earning Tokens through Engagement

Your attendance and engagement are not graded as a separate component. However, I will award **2 additional tokens** (see section E.5) during the quarter to students who demonstrate consistent, active participation. These tokens provide additional retesting opportunities for course standards. See section E.5 for details.

E.2 Weekly Homework

Homework problems will be assigned via Canvas **after every lecture by 6 PM** on that day. Friday, Monday, and Wednesday homework will be **due the following Monday at 11:59 PM**. Specifically,

- the homework problems assigned on Week 1 Monday and Week 1 Wednesday will be due on Week 2 Monday.
- the homework problems assigned on Week 1 Friday, Week 2 Monday, and Week 2 Wednesday will be due on Week 3 Monday.
- Week 8 Friday, Week 9 Monday, and Week 9 Wednesday homework will be due on Week 9 Friday (with up to three days of allowed extension).

Please start working on the homework as soon as it is uploaded and utilize the office hours as necessary; do not wait until the due date to begin.

The homework problems will be a mix of computation, proof writing, and conceptual questions. Each problem is graded by the course assistant on two categories: **Correctness** and **Communication**. The two scores are added together, giving a total of 0 to 4 per problem.

- **Correctness** (0/1/2) assesses whether the mathematics is correct.
  - **0:** Not attempted, or no meaningful progress toward a correct solution.
  - **1:** Demonstrates understanding of the relevant concepts or techniques, but the solution contains a significant error: a wrong conclusion, a logical gap, a misapplied theorem, or an incorrect computation that changes the answer.
  - **2:** Mathematically correct and complete. Minor slips (a sign error in an intermediate step that does not affect the logic, a trivial arithmetic mistake) are acceptable at this level.
- **Communication** (0/1/2) assesses whether the solution is written as a proper mathematical argument. This includes three things: *structure* (is there a clear logical flow with prose connecting the mathematics?), *precision* (are quantifiers stated, notation defined, and mathematical terms used correctly?), and *justification* (is each non-obvious step supported by a stated reason?).
  - **0:** Blank, illegible, or so disorganized that no clear claim or approach is discernible.
  - **1:** The underlying mathematical content is discernible, but the solution falls short as a mathematical argument: bare equations or symbol strings with no connecting prose, missing quantifiers, undefined or misused notation, incomplete sentences, informal language where precise terminology is expected, or non-obvious steps asserted without justification.
  - **2:** Written in clear, connected prose. Quantifiers are stated, notation is used correctly, and each non-obvious step is supported by a stated reason (a computation, a theorem citation, or an explanation). The argument has a clear logical flow from hypotheses to conclusion.

Your overall homework score is calculated as your total score over all questions across all weekly assignments. An assignment with more questions contributes more toward your overall grade.

Submission Requirements

Submit homework through Gradescope. Homework submissions must adhere to the following format requirements:

- Leave the top and left margins blank so that the grader may use this space for scoring and comments.
- Clearly label and number exercises.
- Write legibly and in complete sentences. Solutions must clearly and completely justify answers to earn credit.
- Upload solutions ordered and rotated correctly.

Assignments that do not conform to the format requirements may not earn credit. You may typeset homework using LaTeX, but neat and legible handwritten solutions are acceptable.

To scan and upload your assignment to Gradescope, use the Gradescope Mobile App or follow the Best Practices for Submitting an Assignment. After you have submitted your assignment, you must assign the scanned pages to the questions in the assignment.

Free Extensions

You have **1 free homework extension** for the quarter extending a single homework deadline by 24 hours with no questions asked. To use one, email me **before** the original deadline. Beyond this extension, late homework will not be accepted.

E.3 Checkpoint Quizzes

Rather than traditional midterms and finals, we will have **four in-class checkpoint quizzes**, each 50 minutes:

Quiz Schedule

- **Checkpoint Quiz 1.** Week 2 Friday (April 3).
- **Checkpoint Quiz 2.** Week 4 Friday (April 17).
- **Checkpoint Quiz 3.** Week 6 Friday (May 1).
- **Checkpoint Quiz 4.** Week 8 Friday (May 15).

Each quiz serves double duty: it introduces **new** standards and provides **retesting** opportunities for all previously assessed standards. If you have already earned M on a standard, you do not need to attempt it again. If you initially struggled with a particular standard, you have the time and opportunity to study, practice, and try again.

In addition to the four checkpoint quizzes, there may be short (approximately 10-minute) in-class quizzes on select days throughout the quarter. These provide additional opportunities to demonstrate EC standards and will be announced in advance.

Grading of Standards

EC standards (Examples & Counterexamples) are graded on a two-level scale:

- **M** (Meets Expectations): The response demonstrates clear understanding of the relevant definitions and produces correct, well-chosen examples or counterexamples.
- **N** (Not Yet): The response contains significant errors or does not engage with the core idea.

P standards (Proof & Application) are graded on a three-level scale:

- **M** (Meets Expectations): The proof or solution is correct, complete, and clearly written.
- **R** (Revision Needed): The work contains a minor error but demonstrates full conceptual understanding. You have the opportunity to submit a written correction via a Google Form within **24 hours** of receiving the graded quiz. If the correction convinces me that you understand the error and how to fix it, the R is upgraded to M. Otherwise, it becomes N.
- **N** (Not Yet): The work contains significant errors, circular reasoning, or does not engage with the core idea.

You will receive helpful feedback on unsatisfactory work, the chance to reflect on the feedback, revise your work, and then reassess your understanding in the next quiz. My hope is that this method of grading will keep you clearly informed as to the expectations of the class and how well you are meeting them, while also removing the (often distracting) elements of linear grading that use total points. If you have questions or concerns at any time, please feel free to discuss them with me.

Once a standard is marked M, it stays at M permanently. Only the highest mark ever achieved on each standard counts toward your final grade. I will maintain a record of your **completed** standards (those with M) and periodically update Canvas.



Warning: You may be used to earning partial credit for showing your work. For computational problems, partial progress has clear value. But a proof is different: if the reasoning contains an error, the conclusion is not established, no matter how much correct work surrounds it. This is why we do not award partial credit on standards. The M/R/N system reflects this reality, and the tradeoff is that you get multiple attempts across the quarter, so a single rough day does not determine your grade.

Missed Quiz

If you believe you will miss a quiz, inform me at least a week in advance, if possible (barring emergencies outside your control). Depending on the reason, I may allow a make-up. In most cases, there will be no make-up, as you will have the opportunity to demonstrate the standard again on subsequent quizzes or through the retesting system.

E.4 Oral Final Examination

During finals week (May 26–29), you will have a **20–30 minute oral examination** with me. This is a holistic assessment of your mathematical understanding and does **not** introduce new standards or provide a path to complete standards not yet at M.

The oral final may include:

- Explaining a proof from the course (from lecture, homework, or a quiz);
- Answering conceptual questions that connect ideas across the quarter;
- Discussing examples, counterexamples, or the relationships between theorems;
- Responding to variations of results you have seen.

The oral final is graded holistically as **Strong**, **Satisfactory**, or **Insufficient**, and contributes to your final grade via the threshold table in section F.1. More details on the content of the oral final will be provided later in the quarter.

E.5 Token System

Each student starts the quarter with **2 tokens** and can earn up to **2 additional tokens** through consistent attendance and active participation during class meetings (see section E.1), for a maximum of **4 tokens**. Tokens can be spent as follows:

- **1 token:** Retest one previously assessed standard during office hours. **You may spend at most 2 tokens in any single week, and at most once per standard per week.**

How Retesting Works

There are no separate retesting sessions. Between quizzes, you may come to office hours and spend a token to retest any standard that has been previously assessed. The format will match the corresponding quiz question type.

Note: Unused tokens at the end of the course have no value toward your grade. The 2 participation-based tokens will be awarded at approximately week 4 and week 7 of the quarter.

§F. Assessment and Grading

Your course grade is determined by three components: mastery of course standards, homework performance, and the oral final. The requirements for each grade are outlined below.

F.1 Threshold Grade Requirements

Your final grade is determined by a **checklist**, not a weighted average. To earn a given grade, you must meet **all** requirements in the corresponding column. Your base grade is the highest column for which every requirement is met.

Category	A	B	C	C–	D
EC Standards at M (out of 3)	3	2	2	2	1
P Standards at M (out of 9)	At least 7, including all 5 core	At least 6, including at least 4 core	At least 5, including at least 3 core	At least 5, including at least 2 core	At least 5
Homework	At least 80% of maximum possible	At least 75% of maximum possible	At least 70% of maximum possible	At least 65% of maximum possible	At least 60% of maximum possible
Oral Final	Strong	Satisfactory	Satisfactory	(none)	(none)

Note: If you do not meet the **D** requirements in every category, the grade is an **F**.

Your base grade may be adjusted by $+/-$ based on how close you are to the next higher letter grade. For example, a student with all 4 EC at M, 10 P standards at M (including 4 core), 77% homework, and a satisfactory oral might earn a **B**+. A student with all 4 EC at M, 10 P standards at M (all 5 core), 78% homework, and a satisfactory oral might earn an **A**−. Please contact me at any time during the quarter if you have questions about the grading system or would like to review your current progress.

The final decision regarding any changes to these guidelines will be made by the Director and Co-Directors of Undergraduate Studies in the Department of Mathematics and will be communicated to all via Canvas. Any such changes can only (if anything) loosen the requirements from what is given above.

§G. Collaboration Policy

Collaboration and cooperation are extremely helpful in the learning process. However, it is often unclear what exactly "collaboration" means when working on assignments. The following section should clarify my expectations and give guidelines for avoiding plagiarism.

- **Weekly Homework:** On weekly homework problems, you are permitted to discuss big ideas and hints with your classmates, but every step of every solution should be one that you understand yourself and that you have generated on your own.

  Any collaboration should occur only when your collaborator is at essentially the same stage of the problem solution as you. In particular, if you have not yet started problem #4 and you ask a friend (who has already completed it), "How did you do problem 4?", this counts as **a violation of the university's academic integrity policy**.

  On your written homework, you must indicate who your collaborators are. (If you collaborate with different people on different problems, just say so!)

- **Checkpoint Quizzes:** These must be completed as individual assignments, and collaboration is not permitted. No technology of any kind is allowed.

- **Oral Final:** This is an individual assessment. No collaboration or outside resources.

- **Outside resources:** Unless explicitly permitted, copying outside materials such as solutions found online, web pages, videos, etc., is considered **a violation of the university's academic integrity policy**.

- **Past students or Peer tutors:** On any assignment, basing your work on the efforts of another student who previously completed this course or one like it, or a tutor not specifically approved by me, is considered **a violation of the university's academic integrity policy**.

- **Mathematical tools and software:** Unless otherwise specified, you may use a calculator or mathematical software (such as WolframAlpha or Mathematica) to check your work. You will need to include all steps and details in your submitted work. Neither calculators nor software will be permitted during quizzes.

To ensure productive collaborations, do not work in teams larger than three people on any problem at any given time. In teams larger than that, you run the risk of unequal contribution or social loafing.

G.1 Consequences of academic dishonesty

Violations of academic integrity are serious and will be handled seriously. Resulting punishment could include (at least) taking a zero for an assignment where an instructor has probable cause that cheating or plagiarism has occurred. For more details, regarding academic honesty within the College, please visit the following link: https://college.uchicago.edu/advising/academic-integrity-student-conduct.

G.2 A positive note

Remember that I want you to be successful. That is, I want you to develop a deep, personal understanding of the material we study so that you become a better student of mathematics who can go on to do well in all of your future endeavors. Every part of

this course structure, including both collaborative work and restrictions on collaboration, is intended to help you with this. You will often struggle, and that's intentional. Struggle (and eventual success!) is essential to learning.

In all aspects of the course, please understand that I am generous with hints and am always willing to discuss problems with you. I will never simply give you an answer, but I will offer direction and guidance that will assist you in coming up with a solution on your own. This is by far the most satisfying way to solve a problem, and the difficulty is well worth it. You are always welcome to discuss your questions or concerns with me at any time.

§H. Classroom Norms

H.1 `Growth', not `Ability'

There is a very prevalent belief that you are either "good" or "bad" at math, and if you are "bad" at it, then you will always be bad at it no matter how hard you try. This is extremely false, and the mathematics community bears a lot of responsibility for perpetuating this myth. In reality, mathematics is just like any other discipline or skill: you can improve more and more with practice.

We are all capable of growth in mathematics. You should measure your success in this class by how much your understanding of the concepts has improved over the course of the quarter. Also, math is very hard, so you should expect to struggle with the material! When you struggle, you are learning and growing. Not all people show their struggles in equal ways, so you should always be wary of judging your progress based on your perception of your peers' struggles. You are probably doing better than you think.

H.2 Respecting Each Other

We are not all coming to this class with the same privileges, resources, time, and knowledge. It's really important to keep this in mind when working with each other on homework assignments and during class meetings. **It is our strong belief that as a community, mathematicians and scientists need to do a much better job of making our disciplines more accessible to people of all races, genders (including gender non-conforming folks), sexual identities, and class backgrounds.** While this is a priority for us in the classroom, we do not claim to know how to best honor this commitment, but we are eager to listen, adapt, and learn. So **we are very open to feedback from students when it comes to making the course more accessible and inclusive to all identities.**

It's also important to think about how to respect one another when working together in groups. It's not equally easy for all of us to speak up in a large group, and the voices of historically underrepresented/marginalized students are most easily drowned out in group work. So please keep this in mind when working together. Here are some concrete examples of positive collaborative behavior:

- Making sure everyone who wants it has the opportunity to speak frequently. This can mean checking in with each other to make sure everyone is following along and contributing when they have an idea.
- Respecting people's pronouns and other aspects of their identity.
- Making sure that everyone's ideas are acknowledged when writing up the final solution to a problem. When working in groups, solutions often evolve organically; an idea might pop into your head, and you may think it's yours and yours alone, but perhaps you only arrived there because of something else that someone already said. Pay attention to what people are saying and try to learn from one another.

We will do our best to check in with folks periodically during the quarter. If at any time in the quarter, you find yourself in a group of students for which the above behaviors aren't being practiced and people aren't feeling respected, please let me know.

§I. Helpful Resources

I.1 Office Hours

Office hours are really Student hours! They are an opportunity for you to stop by your instructor's office and ask questions. I have specifically set this time aside in order to give personalized help to individual students. Any and all questions are welcome,

whether you are working on a homework question, have questions on a broader concept discussed in class, or potentially even questions beyond the scope of the course.

Typically, students' misunderstandings in mathematics are initially small, but due to the constructive nature of math, a small misunderstanding can compound as concepts are utilized together. Students who frequently attend office hours and actively engage by asking questions are able to continually resolve their small misunderstandings, resulting in sustained improvement.

Note: College Core Tutors are not permitted to assist with Math 163 per guidance from the MATH department.

Email Responses

I do my best to reply to emails promptly and helpfully. However, I receive a lot of email. To help both you and me, here are some specific expectations about emails:

- If you email me between 8:00 AM and 5:00 PM on a weekday, I'll try to reply to you on the same day.
- If you email me in the evening or overnight (after 5:00 PM), I will try to reply to you the next weekday.
- If your email asks a question that is answered in the Syllabus or on Canvas (such as in an announcement or an assignment sheet), I may reply by directing you to read the appropriate document.

I.2 Academic Accommodations

If you require any special academic accommodations, please provide me with a copy of your Accommodation Determination Letter (issued by the Student Disability Services office) as soon as possible, so we can discuss how your accommodations may be implemented in this course. If you are in the process of obtaining accommodations, please inform me as soon as possible. More information can be found here: https://disabilities.uchicago.edu/.

I.3 Religious Accommodations

The University of Chicago is home to students of all the world's major religions and, though firmly a secular institution, values the rich diversity of spiritual expression and practice found on campus. It is therefore the policy of the University that students who miss class, assignments, or exams to observe a religious holiday must be accommodated as follows: (i) absences may not be counted as a missed class in any course in which attendance is a measure of academic performance; (ii) reasonable extensions of time must be given, without academic penalty, for missed assignments; and (iii) exams must be reasonably rescheduled without academic penalty. You must inform me in writing of your need to observe a religious holiday reasonably well in advance of the absence, preferably at the beginning of the quarter. More information can be found at the following: https://provost.uchicago.edu/handbook/clause/policy-religious-accommodation-missed-classes-assignments-and-exams.

I.4 Wellness Resources

UChicago has counseling available both 24/7 and by appointment through http://wellness.uchicago.edu. Additionally, you can access medical care, including 24/7 support from medical professionals to address your healthcare questions.

I.5 Sexual Misconduct Policy

The University of Chicago recognizes that members of the university community are responsible for ensuring that the community is free from discrimination and other forms of sexual misconduct based on sex or gender, including sexual harassment, sexual assault, stalking, domestic violence, and dating violence. Faculty are considered "Individuals with Title IX Reporting Responsibilities" of the University and are obligated to report information to the Title IX Coordinator related to sexual misconduct. If you think your rights, or the rights of someone else in the university community, have been violated, you can find information on resources and reporting at: https://cares.uchicago.edu/.

Title IX Coordinator: Bridget Collier, Associate Provost & Director (bcollier@uchicago.edu, 773-702-5671)

§J. Recording and Deletion Policy

The Recording and Deletion Policies for the current academic year can be found in the Student Manual under Petitions, Audio & Video Recording on Campus.

- Do not record, share, or disseminate any course sessions, videos, transcripts, audio, or chats.
- Do not share links for the course with those not currently enrolled.
- Any Zoom cloud recordings will be automatically deleted 90 days after the completion of the recording.

§K. Disclaimer

I reserve the right to make changes to this syllabus if needed. Any changes will be announced to the class in a timely manner.

§L. Math 16300 Course Standards

There are **12 standards** in two categories: 3 EC standards (Examples & Counterexamples) and 9 P standards (Proof & Application). EC standards are graded **M/N**. P standards are graded **M/R/N**. See the Assessment section for details on each grade level. The five ⍟ marked standards are considered the most important (core) standards. See section F.1 for how these factor into your final grade.

L.1 EC Standards (Examples & Counterexamples)

EC01. **Sequences.** I can **construct** sequences with prescribed properties (e.g., bounded but divergent, monotone with a given limit, Cauchy), **identify** which properties a given sequence satisfies, and **produce** counterexamples to false claims about sequences.

EC02. **Series.** I can **give** examples of series that satisfy specified convergence behavior, and **produce** counterexamples to false claims about series.

EC03. **Uniform convergence.** I can **construct** function sequences with specified convergence behavior (e.g., pointwise but not uniform convergence, pointwise convergence with failure of term-by-term integration) and **identify** the mode of convergence on a given domain.

L.2 P Standards (Proof & Application)

P01. ⍟ I can **write** complete ε–N proofs of convergence and divergence from the definition or by combining hypotheses, using techniques such as the $\varepsilon/2$ or the $\max\{N_1, N_2\}$ argument, the triangle inequality, and algebraic identities to control the relevant quantities.

P02. I can **apply** the Monotone Convergence Theorem by verifying monotonicity, establishing a bound, concluding convergence, and determining the limit.

P03. ⍟I can **prove** that a sequence converges when the limit is not known in advance, using the Bolzano–Weierstrass theorem or the Cauchy criterion. I can also determine the limit when applicable.

P04. I can **select** and **apply** the appropriate convergence test (comparison, integral, ratio, root) to determine convergence or divergence of a nonnegative series.

P05. ⍟ I can **classify** a series as absolutely convergent, conditionally convergent, or divergent, and **apply** the alternating series test or Dirichlet's test as appropriate.

P06. I can **prove** or **disprove** that a given function sequence converges uniformly on a specified domain using sup-norm estimates or explicit constructions, and **apply** the appropriate uniform limit theorem to justify interchange of limits with continuity, integration, or differentiation.

P07. ⍟ I can **establish** uniform convergence of a series of functions using the Weierstrass M-test or other methods, and **apply** the series corollaries to justify term-by-term integration, differentiation, or continuity of the sum.

P08. ⍟ I can **determine** the interval of convergence of a power series (radius and endpoint behavior) and **justify** term-by-term differentiation, integration, and algebraic operations within the radius.

P09. I can **derive** the Taylor series of a given function by substitution, term-by-term differentiation or integration, or algebraic manipulation of known power series, and **state** the interval of validity.